1 Introduction

This laboratory session aims to learn the application of Hall effect sensors in conjunction with DC motors equipped with encoders. Through the use of these encoders, we will investigate the characteristics and implications of signal variation as a function of the accelerated motor speed. The experiments are providing an understanding of the encoder's functionality and its responsiveness to changes in the magnetic field through practical application.

2 Experiments

In this laboratory session, we will conduct two experiments. The first will focus on visualizing the encoder output data by connecting the DC motor to various voltage levels and observing the resulting changes in encoder output on an oscilloscope. The second experiment will explore calculating the speed of the DC motor using Arduino and interrupts to measure the frequency of the encoder's output signals. These experiments are expected to illustrate the fundamental principles of motor speed control and feedback mechanisms through Hall effect sensors.

2.1 Experiment 1: Implementing button

In the first part of our experiments, we constructed the setup shown in the Figure 1 to observe the output from a DC motor encoder under varying voltage conditions. The configuration included the following components:

- Arduino
- Oscilloscope
- Jumper Cables
- 1x DC motor with encoder

And the cable connections should done with these specializations below.

- Red: + motor .....(to power supply +)
- Black: - motor ..... (to power supply -)
- Green: Encoder GND .... (to Arduino GND)
- Blue: Encoder Vcc .... (to Arduino 5v)
- Yellow: HALL Effect A Vout ... (Connected to Oscilloscope to read encoder data)
- White: HALL Effect B Vout ... (not connected)

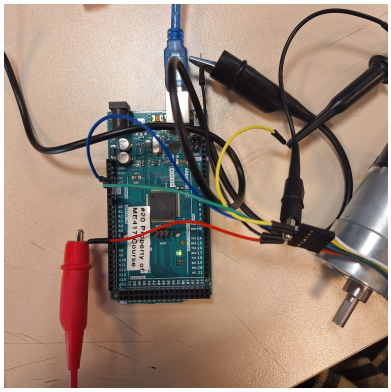


Figure 1: Arduino push-button connection schematic.

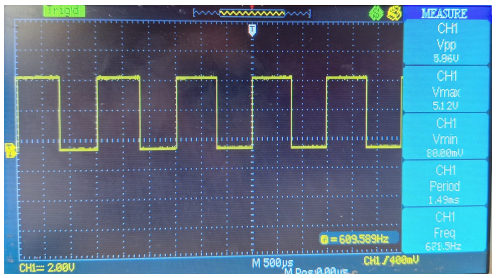


Figure 2: Oscilloscope screen while applying 3Volts to motor. Measuring 675 Hz.

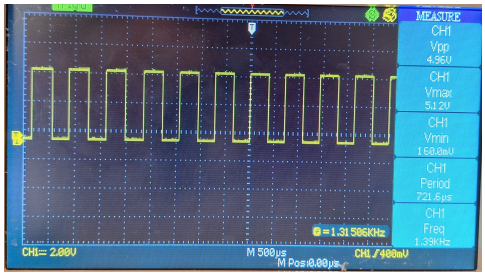


Figure 3: Oscilloscope screen while applying 6Volts to motor. Measuring 1.39 KHz.

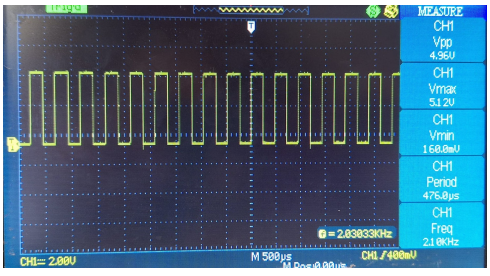


Figure 4: Oscilloscope screen while applying 9Volts to motor. Measuring 2.1 KHz

We began the experiment by connecting the DC motor to the power supply set at 3 Volts through the Black and Red cables, as specified in the wiring guide. This setup was aimed at feeding the Hall effect sensor circuit through Arduino's 5V output. Observations were made on the oscilloscope to monitor the encoder's square wave outputs corresponding to the motor's speed.

The oscilloscope displayed the encoder's output when the DC motor was supplied with 3 Volts. As shown in Figure 2, the square wave output at this voltage showed a lower frequency, which is our reference for other measurements. As we increased the voltage to 6 and then 9 Volts, we observed a clear increase in the frequency of the square waves, as shown in Figure 3 and Figure 4. These observations point to a direct relationship between the supply voltage and the motor's speed, shows the encoder's ability to capture and translate motor speed variations into measurable electrical signals.

In the experiments, we analyzed the behavior of a DC motor equipped with an encoder under different voltage supplies. The angular velocity (ω) and the period (T) of the square wave outputs from the encoder were calculated to understand the motor's performance characteristics. Angular velocity has determined using the formula $\omega = 2\pi f$, where f is the frequency of the encoder output. The period T of the encoder signal, indicative of the time between successive waves, was computed as $T = \frac{1}{f}$.

Measurements were made at three different voltages: 3V, 6V, and 9V, giving frequencies of 675 Hz, 1390 Hz, and 2100 Hz, respectively. These frequencies correspond to specific angular velocities and periods as calculated and presented in the Table 1. This analysis helps to quantify how the motor speed varies with changes in the supply voltage.

Table 1: Calculated Angular Velocities and Periods at Different Voltages

Voltage (V)	Angular Velocity (ω) [rad/s]	Period (T) [s]
3V	$2\pi \times 675 \approx 4241.15$	$\frac{1}{675} \approx 0.00148$
6V	$2\pi \times 1390 \approx 8730.98$	$\frac{1}{1390} \approx 0.00072$
9V	$2\pi \times 2100 \approx 13194.69$	$\frac{1}{2100} \approx 0.000476$

2.2 Experiment 2 - Calculating the Speed of a DC Motor Using Arduino

In this part, we transition from visual observation of the encoder output to a more quantitative analysis using Arduino to calculate the motor speed. This method uses the interrupts feature of the Arduino to accurately and efficiently measure the speed of the DC motor based on the encoder signals.

We began by connecting the output of the encoder to digital pin 2 of the Arduino, configuring it to trigger an interrupt on the rising edge of the signal. This configuration provided us information to count the number of pulses generated by the encoder, which directly correlate with the motor's rotation.

The Arduino code is designed to count the pulses within a given time frame and calculate the revolutions per minute (RPM) of the motor. The encoder we used generates 64 pulses per revolution of the motor shaft. Given the gear ratio of the motor is 18.75:1, the effective pulse count needed to determine a complete revolution of the output shaft was calculated as 1200 pulses per revolution (18.75×64).

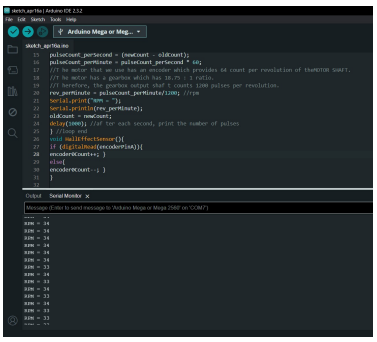
2.3 Code Implementation

```
#define encoderPinA 2 // Pin connected to Hall effect sensor A output
volatile long encoderCount = 0; // Variable to store encoder count

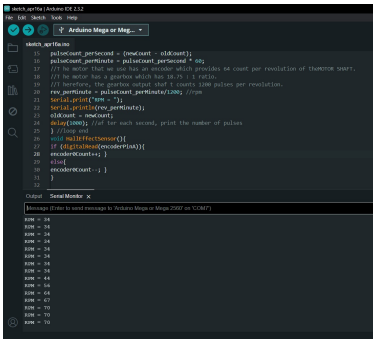
void setup() {
    pinMode(encoderPinA, INPUT_PULLUP);
    attachInterrupt(digitalPinToInterrupt(encoderPinA), countPulse, RISING);
    Serial.begin(9600);
}

void loop() {
    long count = encoderCount; // Read the current count
    float rpm = (count / 1200.0) * 60; // Calculate RPM
    Serial.print("RPM: ");
    Serial.println(rpm);
    encoderCount = 0; // Reset count after each measurement
    delay(1000); // Update rate of 1 second
}

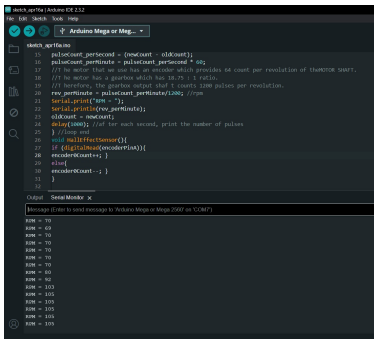
void countPulse() {
    encoderCount++; // Increment the pulse count
}
```



(a) Arduino serial output with 3 Volts power supply, calculations shows 33 34 RPM.



(b) Arduino serial output while changing 3 Volts power supply to 6 Volts, calculations shows 70 RPM.



(c) Arduino serial output, changing 6 Volts power supply to 9 Volts, calculations shows 105 RPM.

Figure 5: Arduino serial outputs of calculation of RPM of the Motor.

2.4 Results and Analysis

The motor's speed was calculated by counting the number of pulses for one-second intervals, translating these pulses into RPM based on the encoder's output and the motor's gearbox ratio. Displayed on the Arduino's serial monitor, the RPM consistently increased with the supply voltage—33 RPM at 3 volts, 70 RPM at 6 volts, and 105 RPM at 9 volts. This clear correlation between voltage and RPM underscores the precision of using interrupts in microcontroller programming for real-time data acquisition. Such capabilities are crucial in applications requiring exact control over mechanical systems, as seen in robotics and automation, where real-time monitoring and control are essential for advanced mechatronic systems.